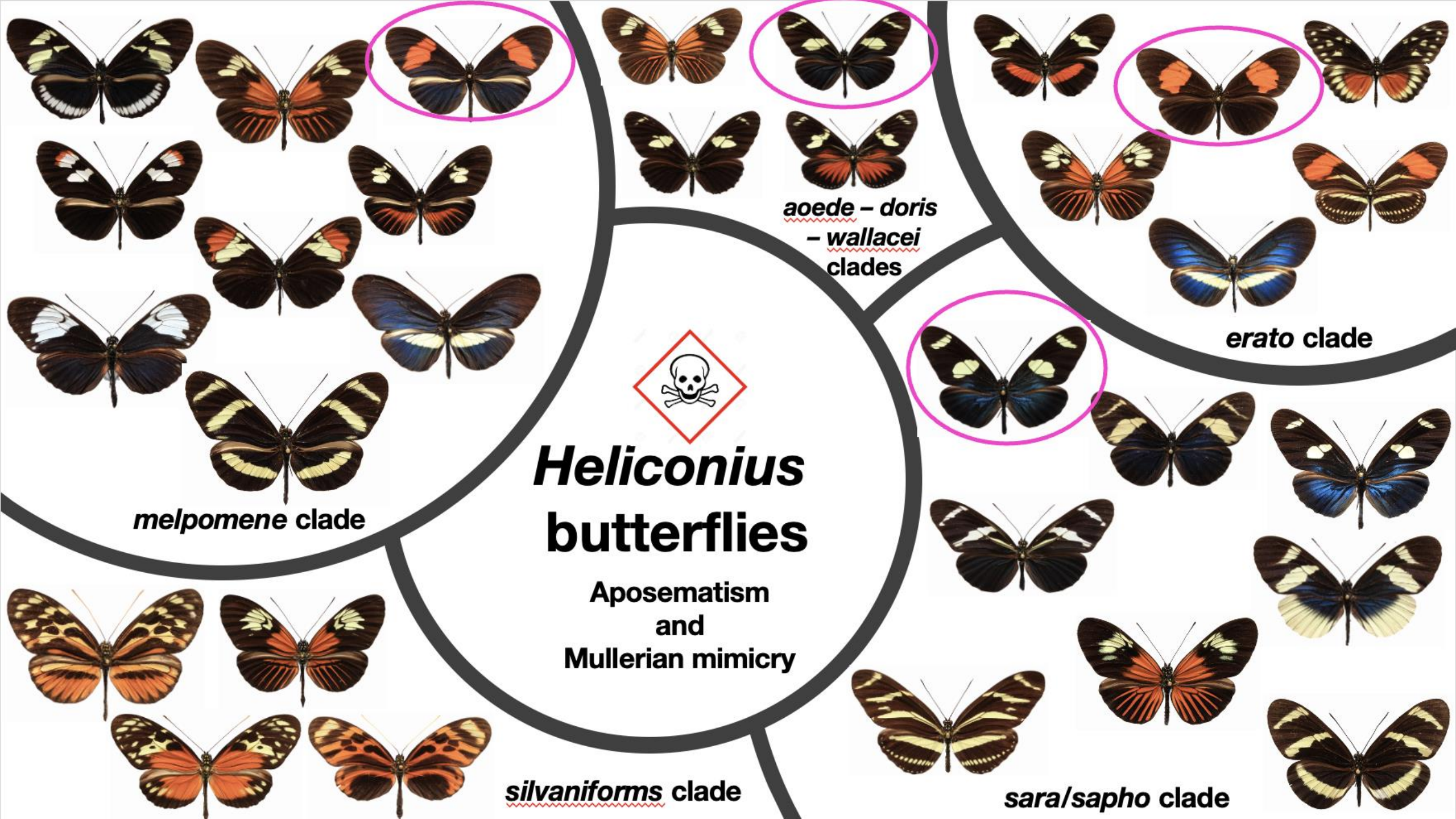
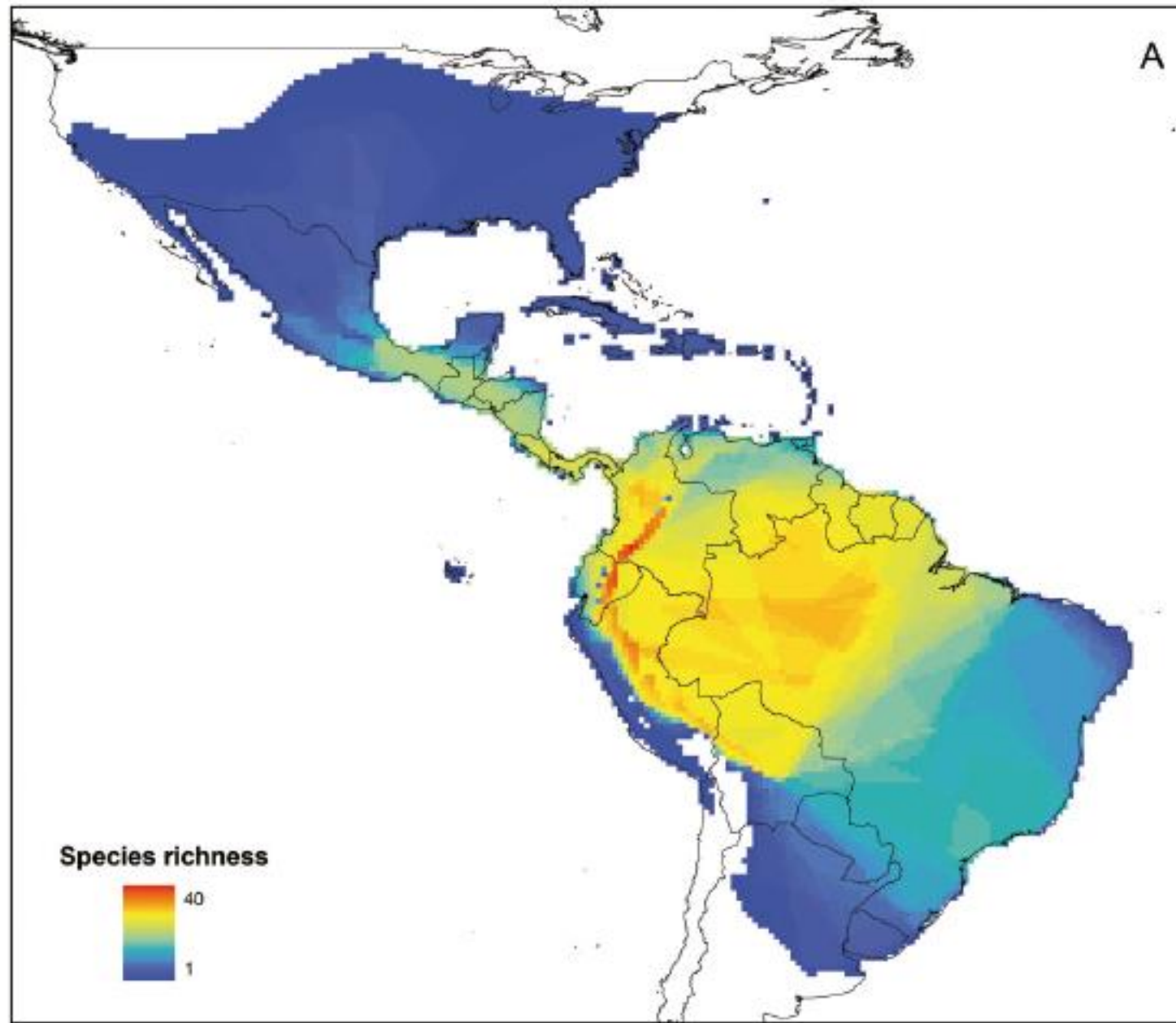


**Introduction to
Heliconius butterflies**

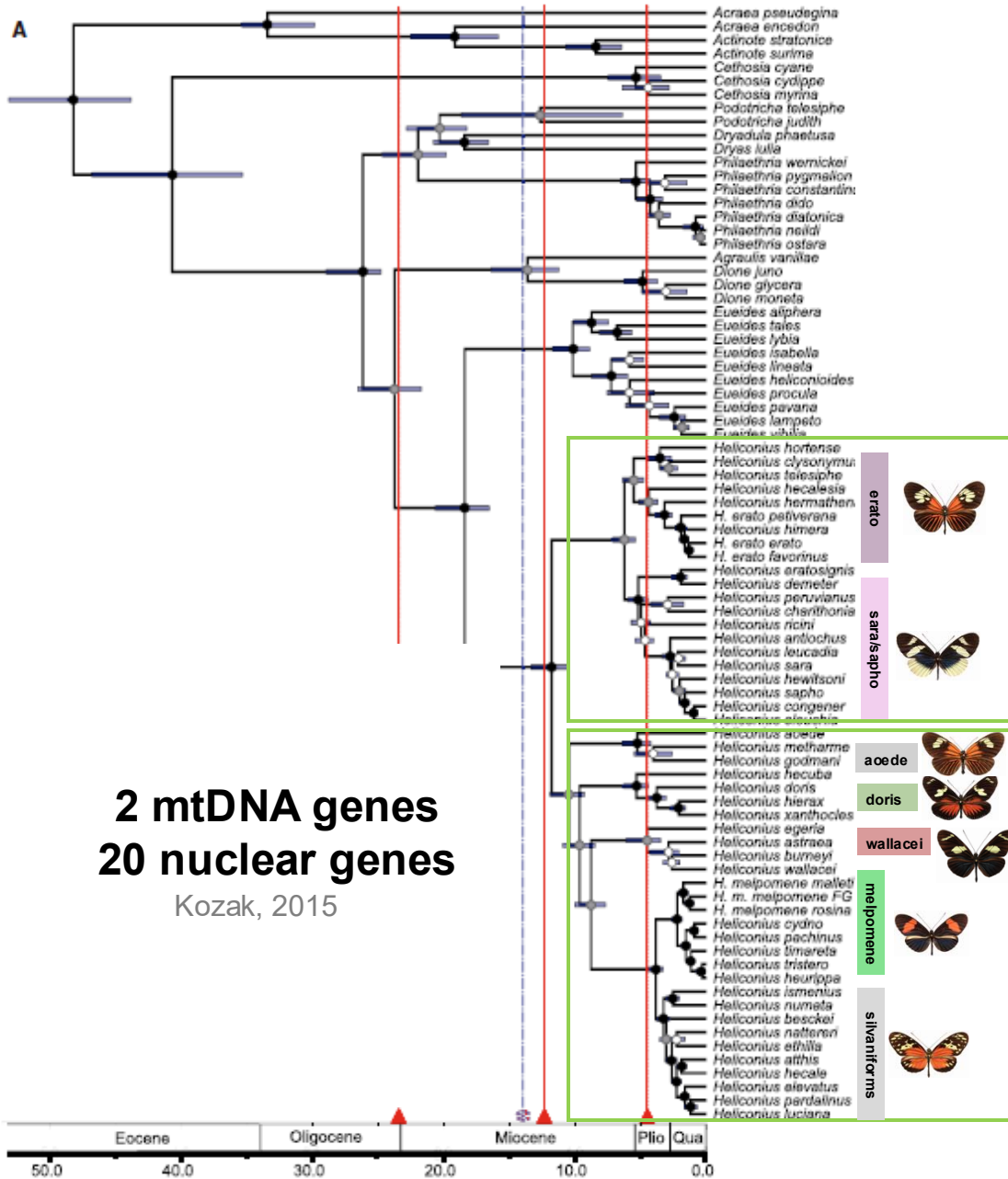
**Biodiversity
genomics course
Mendoza-Argentina
Nov. 2025**





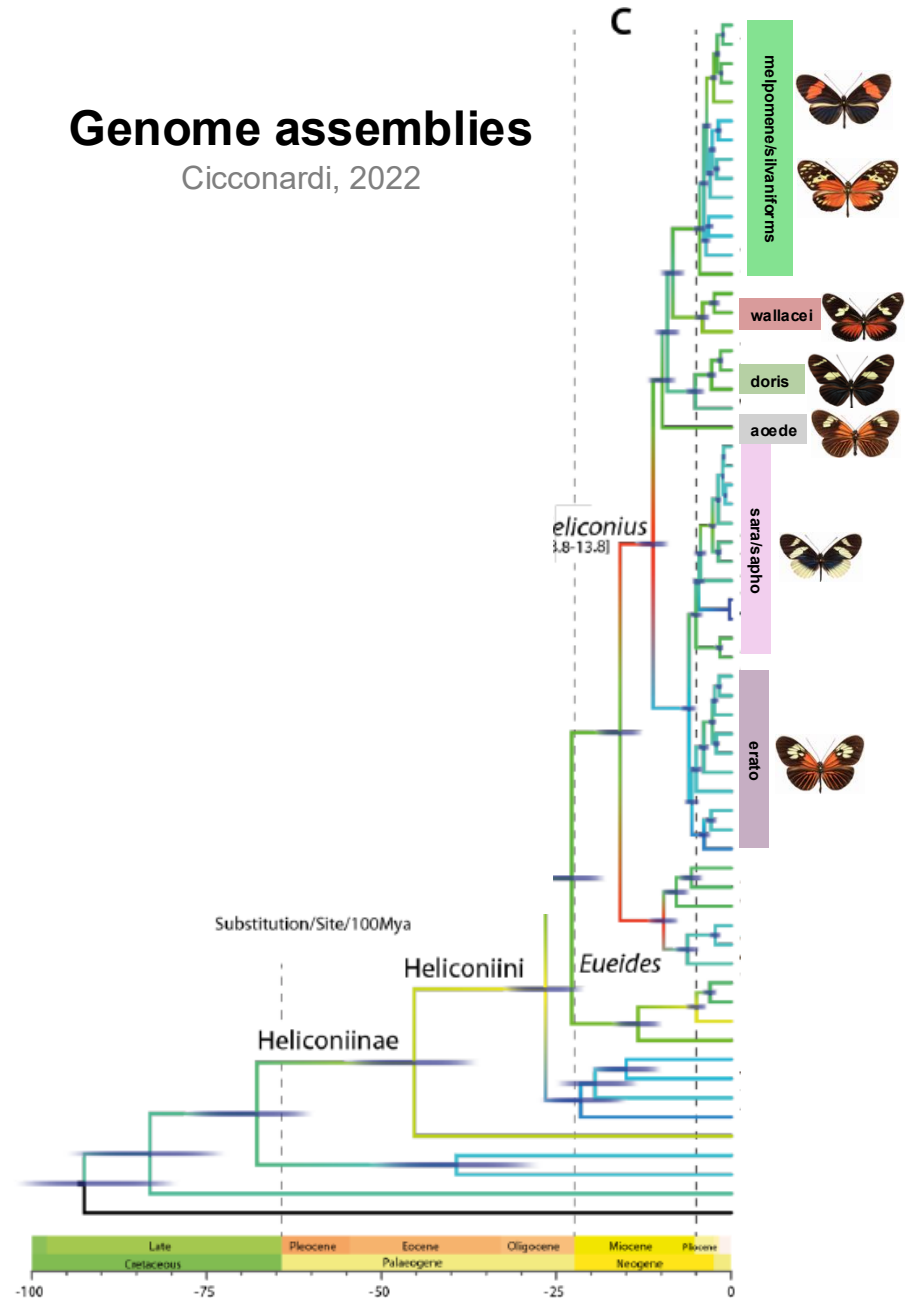


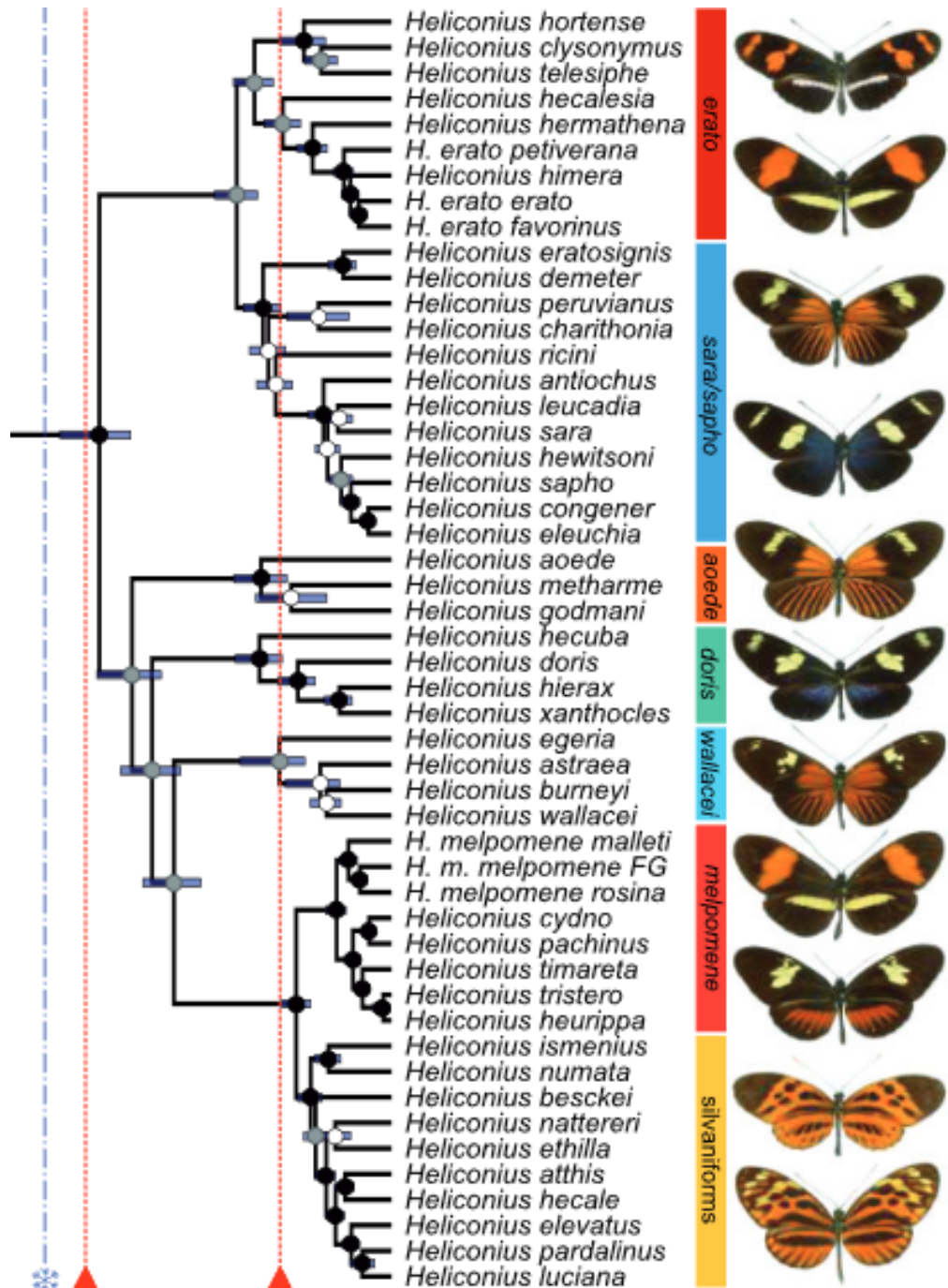
Introduction



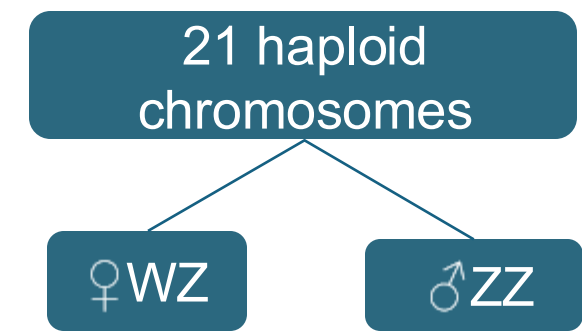
Genome assemblies

Cicconardi, 2022



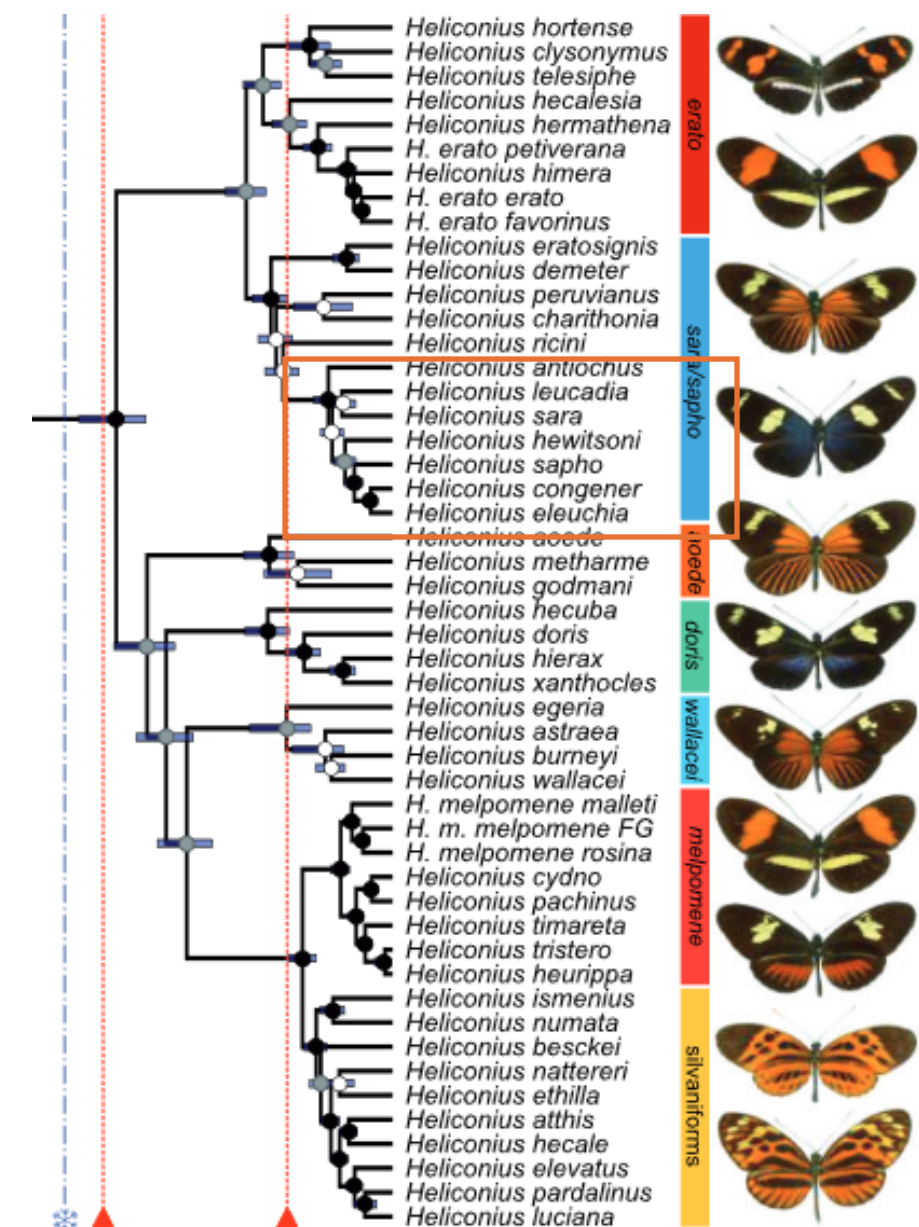


Neotropical *Heliconius* butterflies



Sex chromosome system

Introduction

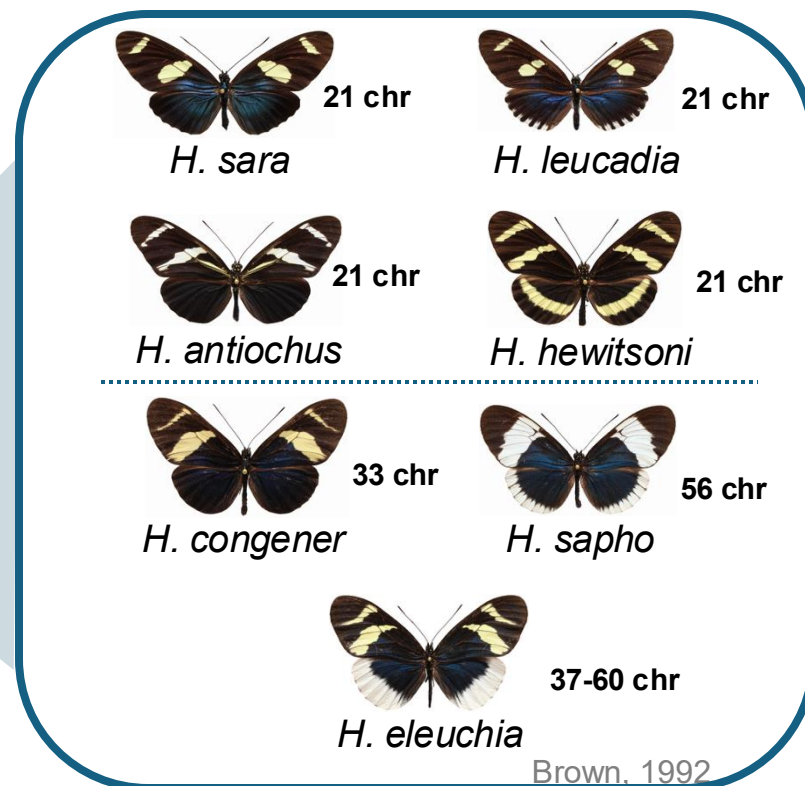


21 - 60
chromosomes

Sex chromosome
system

♀ $W_1W_2W_3Z_1Z_2Z_3$

♂ $ZZ_1ZZ_2ZZ_3$

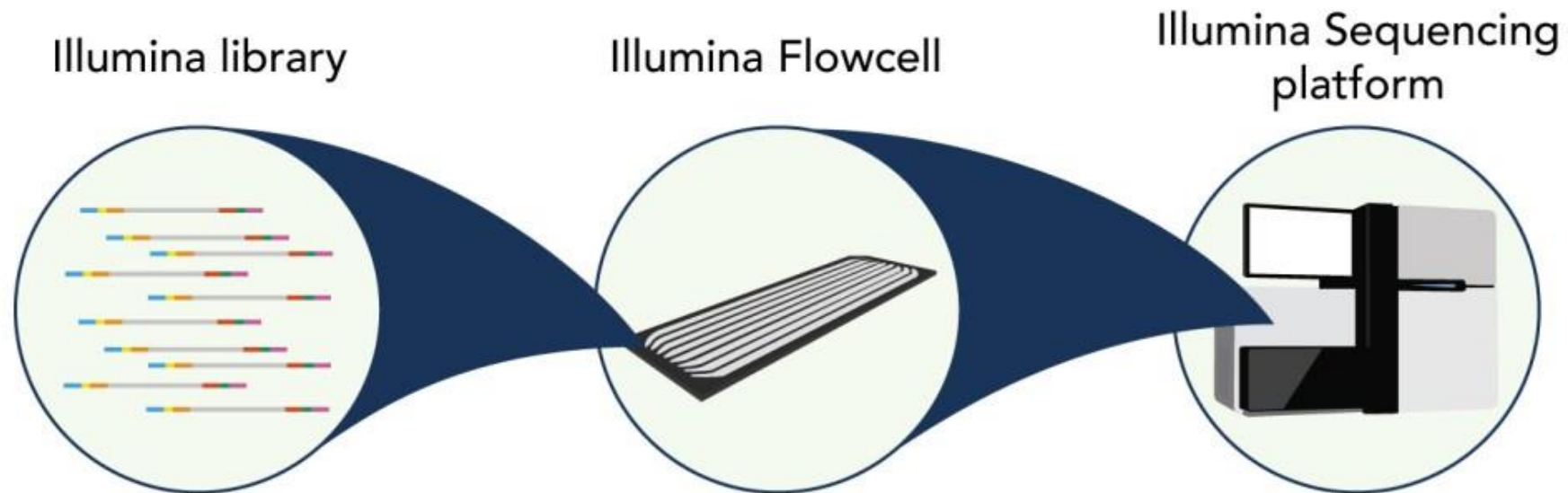


The data (**raw reads**) we
will be working with today
and tomorrow are from this
clade.

How do raw sequences
files look like?

How can we do a quality
control of them?

Raw sequences are store in FASTQ files



<https://joshuapenalba.com/illumina-sequencing/>

A FASTQ file is the standard way to store millions of short DNA fragments, this format include:

- the DNA sequence *and*
- how confident the machine was about each base it read.

FASTQ files extension could be *.fastq* or *.fq*. If compressed should be *.fastq.gz* or *.fq.gz*

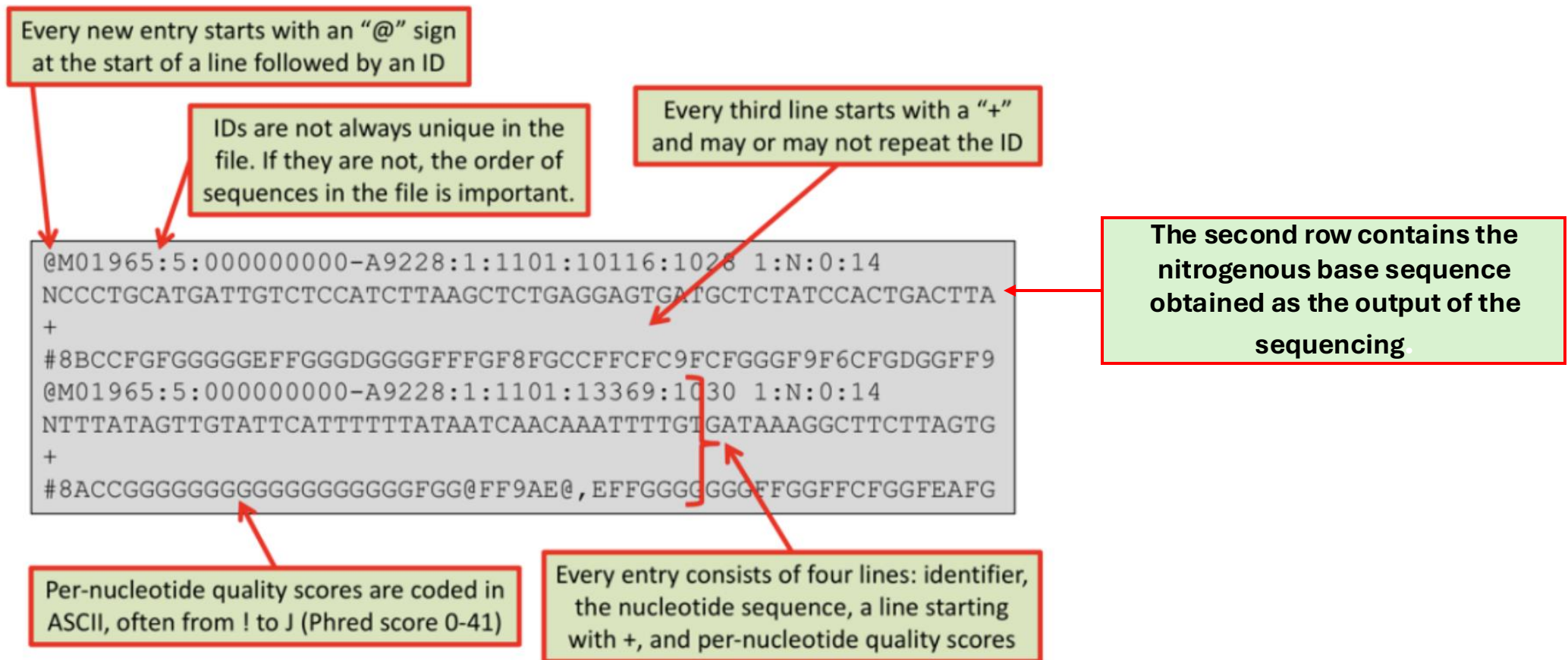


Figure 1. Structure of a FASTQ file. Source: "Theoretical Biology and Bioinformatics" course, Utrecht University, Bas E. Dutilh & Can Keşmir

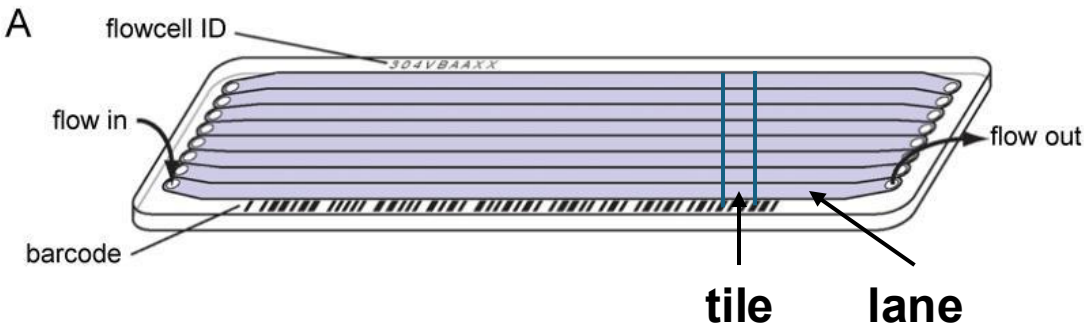
Read header starts with an @ and includes sequencing metadata

Illumina read header format

@A00469:53:H7VFYDSX2:1:1101:1122:1000 1:N:0:AGTCAA

↑ colon

Field	Example	Description
Instrument name	A00469	Sequencer ID
Run number	53	Run identifier
Flowcell ID	H7VFYDSX2	Flowcell barcode
Lane	1	Lane number on flowcell
Tile	1101	Tile within the flowcell
X-coordinate	1122	Cluster X-position
Y-coordinate	1000	Cluster Y-position
Read number	1	Read pair (1 = R1, 2 = R2)
Is filtered	N	Y/N for passing Illumina chastity filter
Control number	0	Internal control indicator
Index sequence	AGTCAA	Sample index/barcode



In paired-end sequencing:

- Read 1 and Read 2 will have headers that **only differ in the read number** (1:N: vs 2:N:).
- Tools like fastp, bowtie2, or bwa use this info to recognize and synchronize pairs.

FASTQ files quality scores

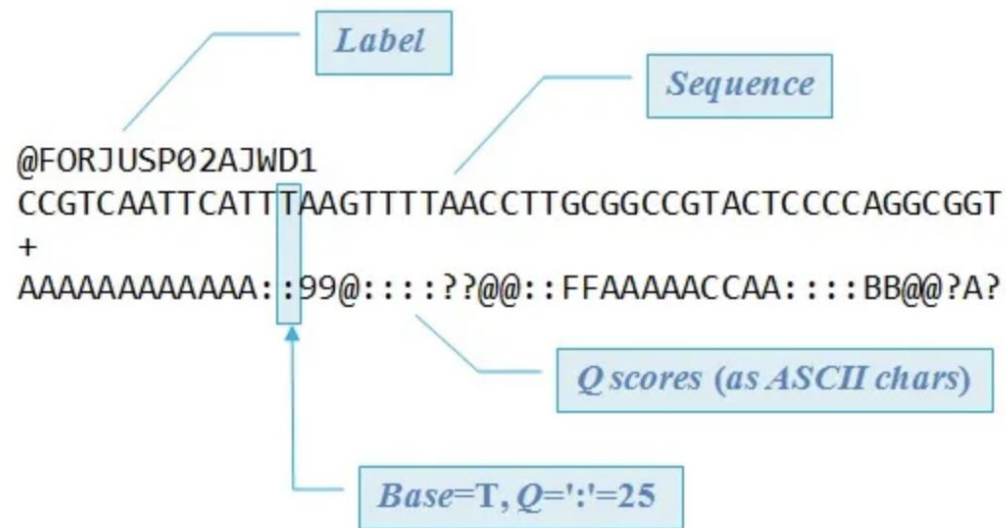


Figure 2. Detailed structure of a FASTQ file.

Source: https://www.drive5.com/usearch/manual/fastq_files.html

Table 1. ASCII characters from ! to J (ASCII values 33-74) encode Phred quality scores from 0 to 41. This corresponds to a probability of 10^0 (100%) to $10^{-4.1}$ (<0.01%) that the nucleotide at that position is wrong. Phred+33 encoding is e.g. used by traditional Sanger sequencing and by the latest Illumina machines.

!	33	0	'	39	6	-	45	12	3	51	18	9	57	24	?	63	30	E	69	36
"	34	1	(	40	7	.	46	13	4	52	19	:	58	25	@	64	31	F	70	37
#	35	2	)	41	8	/	47	14	5	53	20	;	59	26	A	65	32	G	71	38
\$	36	3	*	42	9	0	48	15	6	54	21	<	60	27	B	66	33	H	72	39
%	37	4	+	43	10	1	49	16	7	55	22	=	61	28	C	67	34	I	73	40
&	38	5	,	44	11	2	50	17	8	56	23	>	62	29	D	68	35	J	74	41

ASCII character	ASCII value	Phred score
-----------------	-------------	-------------

Figure 3. ASCII characters. Source: "Theoretical Biology and Bioinformatics" course, Utrecht University of Bas E. Dutilh & Can Keşmir

Usually Phred scores are between **0**, which means an error rate of **100%**, and **41**, which is a probability of **0.01%** that the nucleotide is wrong, and therefore an error.

Transforming the Phred score into ASCII allows us to have only one digit per quality value. This reduces the size of the file and allows a 1:1 match between nucleotide and quality score.

How can we assess quality of the Fastq files?

Why assess quality?

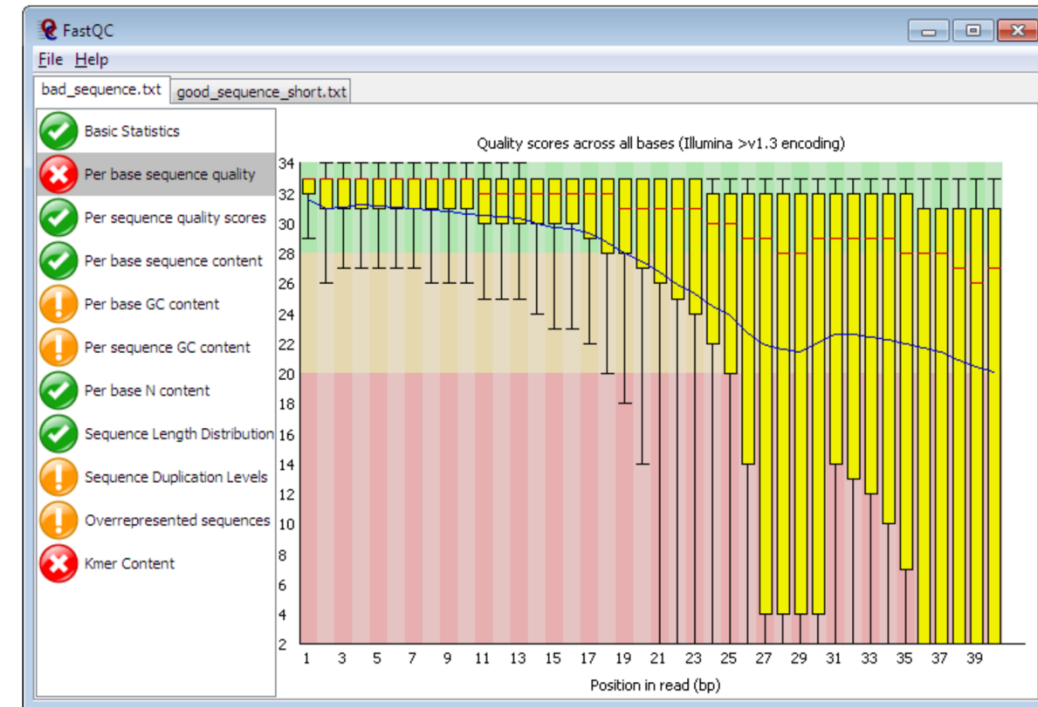
Because raw sequencing data often contains:

- Low-quality bases, especially at read ends
- Adapter contamination
- Overrepresented sequences (PCR artifacts)
- Biased GC content
- Unusual base composition
- Incomplete or failed reads

Poor-quality reads can:

- Bias alignment
- Create false variants

FastQC software



<https://www.bioinformatics.babraham.ac.uk/projects/fastqc/>

```
fastqc -t 4 sample1.fastq.gz
```

OR

```
fastqc -t 4 -o ./fastqc_results/ /path/to/fastqs/*.fastq.gz
```

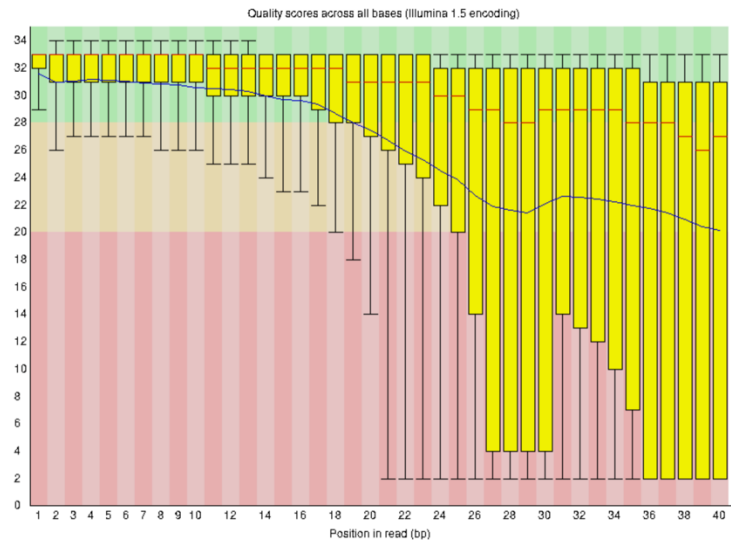
```
cd fastqc_results/  
multiqc .
```

Let's have a look at the first few sequences
and check the sequencing quality with fastqc

Interpreting FastQC results

Bad sequence

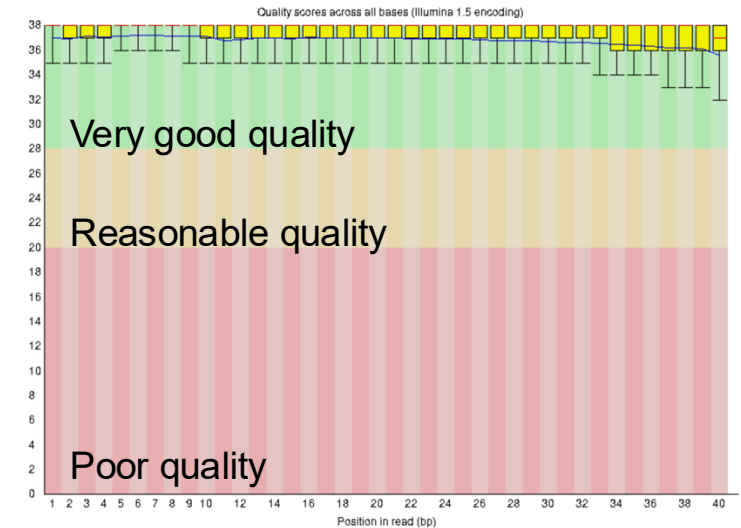
✖ Per base sequence quality



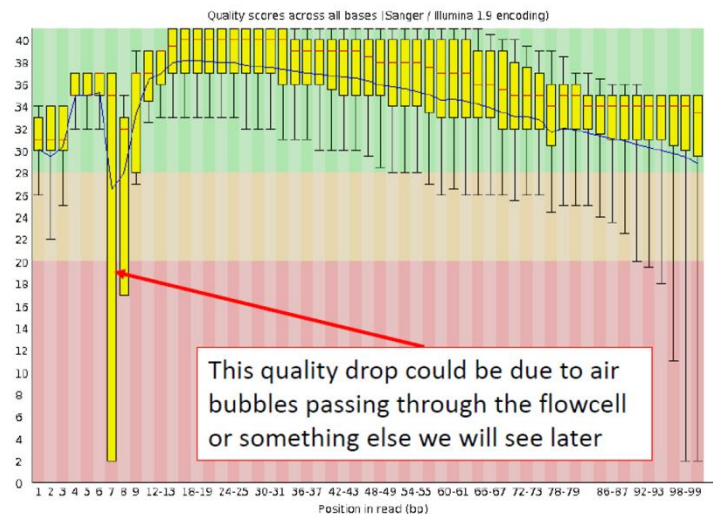
- The higher the score, the better the base call.
- The quality of reads on most platforms will drop at the end of the read. This is often due to signal decay or phasing during the sequencing run.

Good sequence

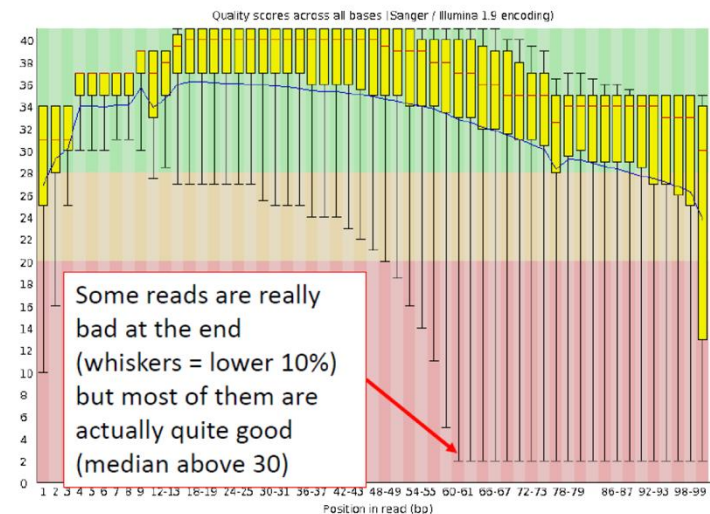
✔ Per base sequence quality



RAD1



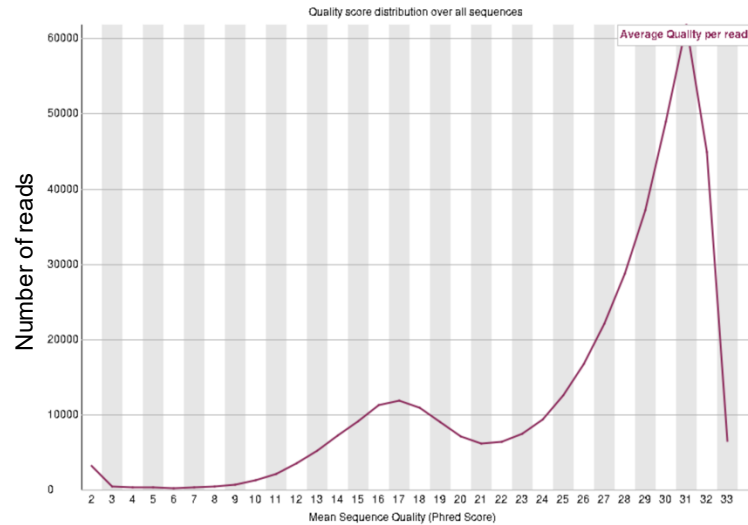
RAD2



Interpreting FastQC results

Bad sequence

✓ Per sequence quality scores

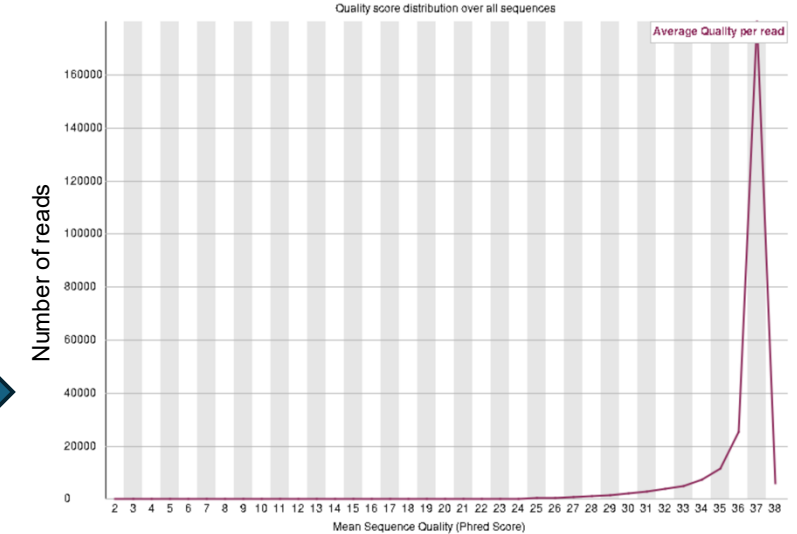


- Indicates **heterogeneous quality**: some reads are very high quality, but a sizable fraction is lower.
- **Trimming** or filtering reads with low mean quality.

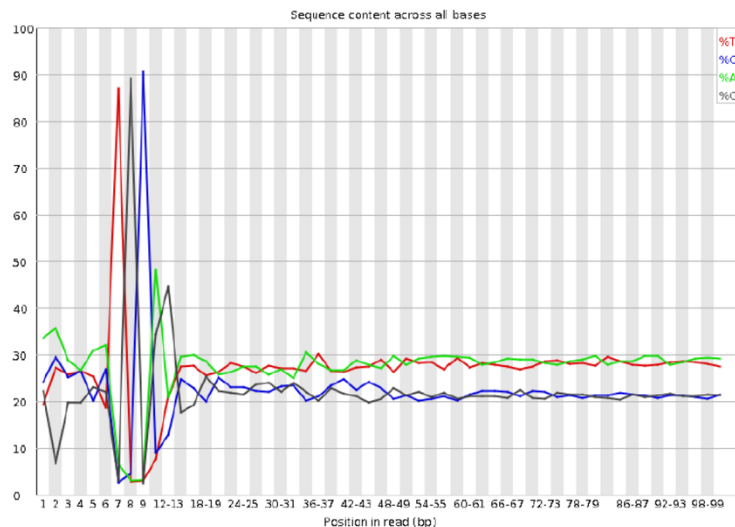
- Almost all reads are very high quality.
- Tight distribution centered around Phred 36–38.

Good sequence

✓ Per sequence quality scores

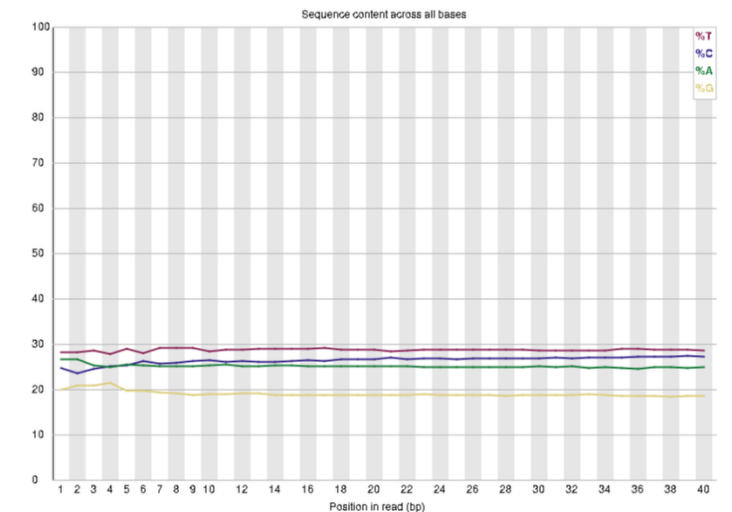


Per base sequence content



- Proportion of each nucleotide (A, T, C, G) at each position in the reads.
- These plots help detect **sequence composition bias**, which can arise from library preparation artifacts, adapter contamination, or non-random fragmentation.

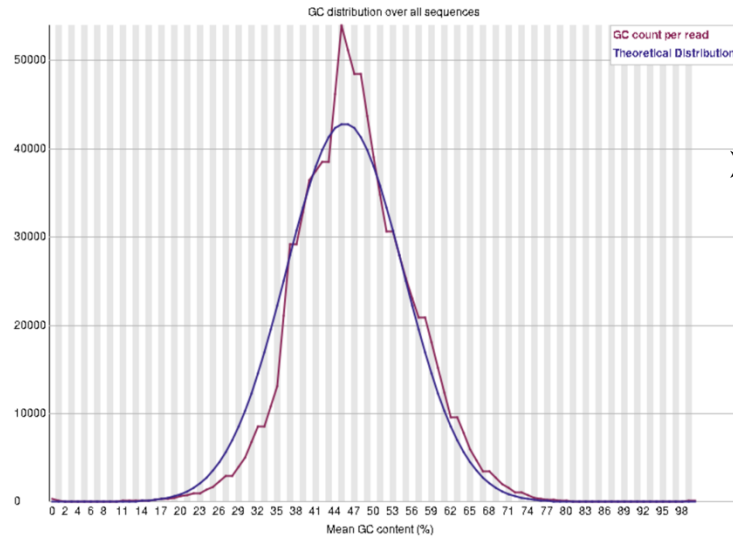
✓ Per base sequence content



Interpreting FastQC results

Bad sequence

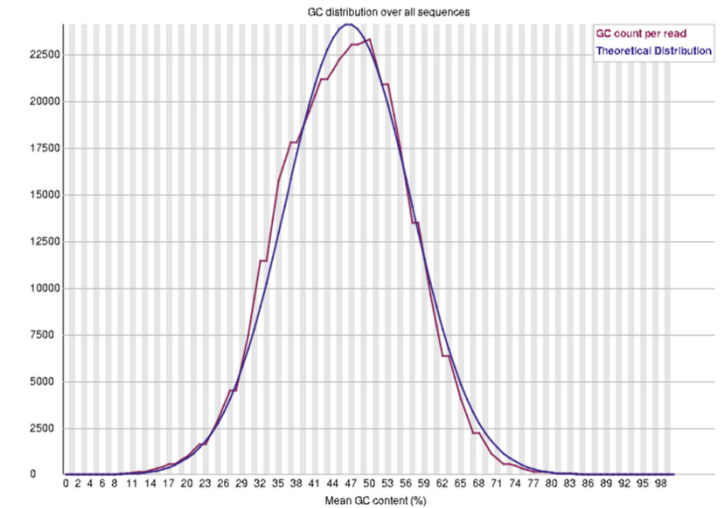
⚠ Per sequence GC content



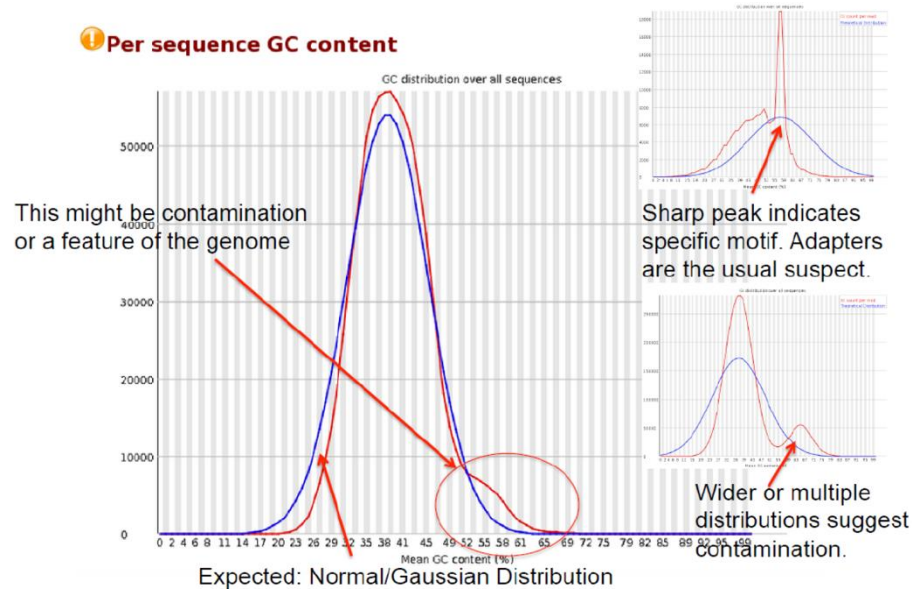
➤ These figures show a comparison between the **observed GC content distribution (red)** of your sequencing reads to an **expected (theoretical) distribution (blue)**.

Good sequence

✅ Per sequence GC content



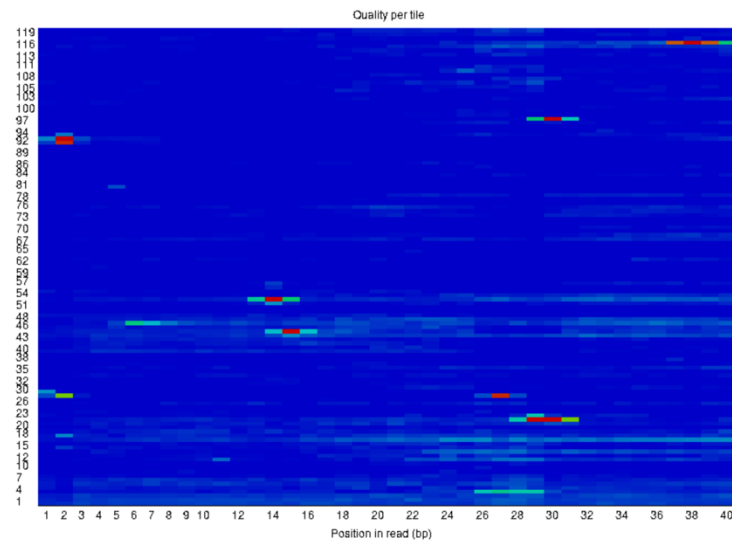
⚠ Per sequence GC content



More examples of what you do not want to see:

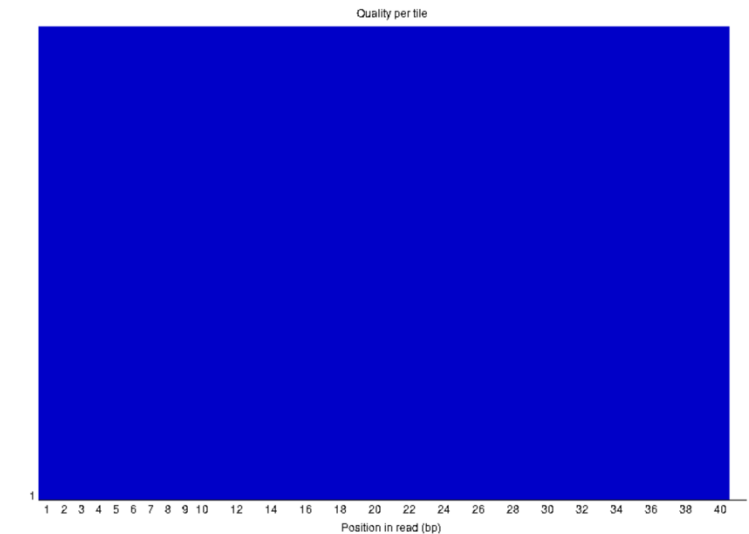
Interpreting FastQC results

✗ Per tile sequence quality

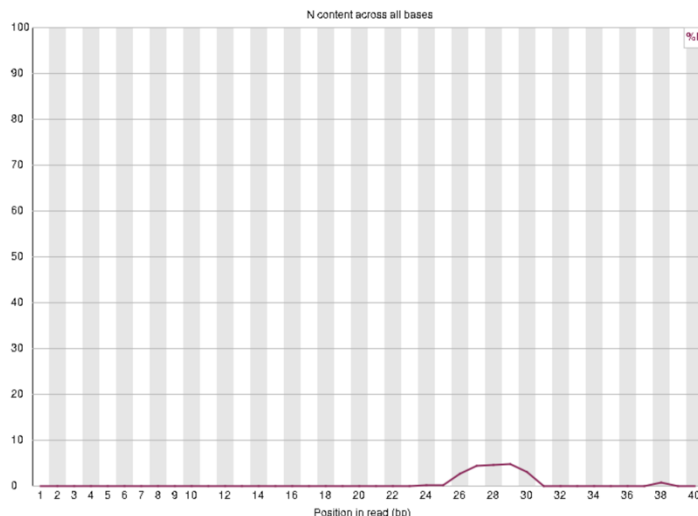


- Quality scores from each tile across all of your bases to see if there was a loss in quality associated with only one part of the flowcell.
- The hotter colours indicate that reads in the given tile have worse qualities for that position than reads in other tiles.

✓ Per tile sequence quality

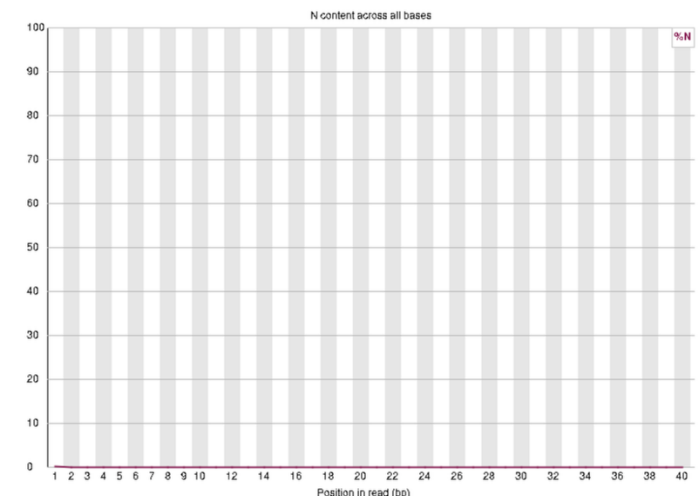


✓ Per base N content



- This figure reports the **percentage of ambiguous bases ("N")** at each position along the reads.
- This can result from:
 - Low signal intensity
 - Mixed signals
 - Sequencing chemistry failure
 - Base-calling uncertainty

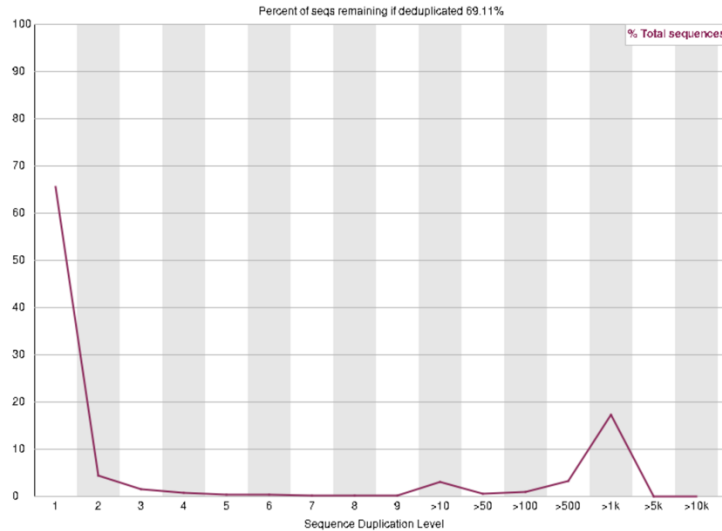
✓ Per base N content



Interpreting FastQC results

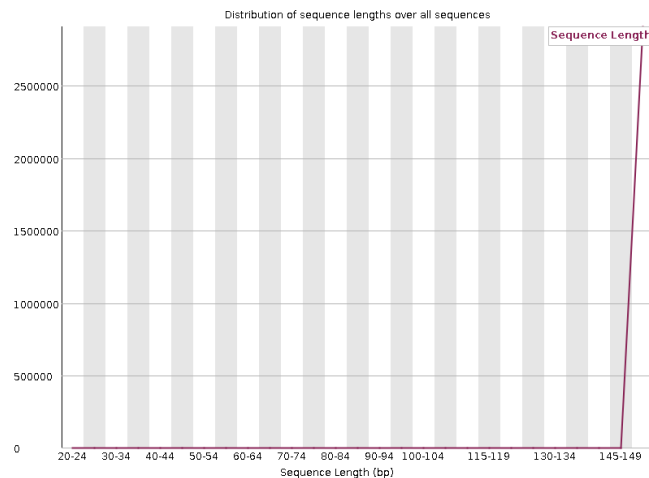
Bad sequence

⚠ Sequence Duplication Levels



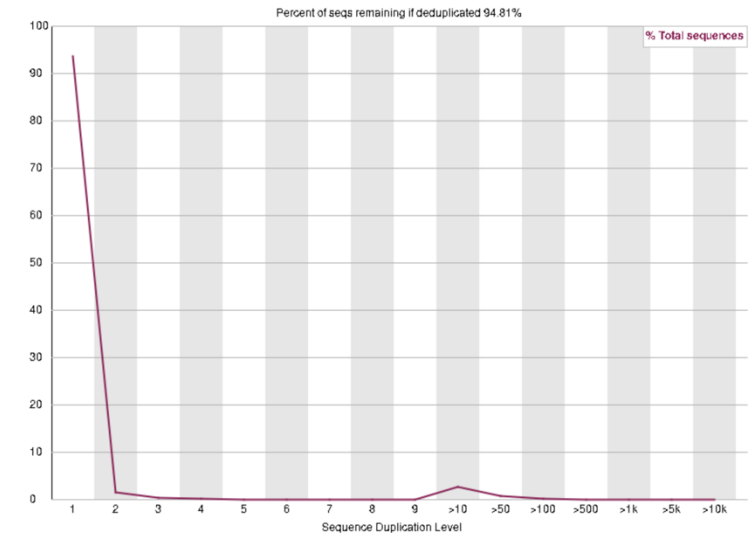
- This figure shows how many **duplicate sequences** exist in your data.
- A lot of duplication could mean:
 - Overamplification during PCR
 - Low input sample
 - Technical artifact, such as adapter dimers or contamination

Sequence length distribution



Good sequence

✅ Sequence Duplication Levels



- This plot shows the distribution of fragment sizes in the file which was analysed.
- In many cases, this will produce a simple plot with a peak at one size. But some sequencers could produce reads of widely varying lengths.